



Atezolizumab plus bevacizumab and chemotherapy in non-small-cell lung cancer (IMpower150): key subgroup analyses of patients with *EGFR* mutations or baseline liver metastases in a randomised, open-label phase 3 trial

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Summary

Background The IMpower150 trial showed significant improvements in progression-free and overall survival with atezolizumab plus bevacizumab plus carboplatin plus paclitaxel (ABCP) versus the standard-of-care bevacizumab plus carboplatin plus paclitaxel (BCP) in chemotherapy-naïve patients with non-squamous non-small-cell lung cancer. Here, we report the efficacy of ABCP or atezolizumab plus carboplatin plus paclitaxel (ACP) versus BCP in key patient subgroups.

Methods IMpower150 was a randomised, open-label, phase 3 study done at 240 academic medical centres and community oncology practices across 26 countries worldwide. Patients with chemotherapy-naïve metastatic non-small-cell lung cancer were randomly assigned (1:1:1) to receive ABCP, ACP, or BCP every three weeks. The co-primary endpoints were overall survival and investigator-assessed progression-free survival in intention-to-treat wild-type patients (patients with epidermal growth factor receptor [*EGFR*] or anaplastic lymphoma kinase [*ALK*] genetic alterations were excluded). Efficacy was assessed in key subgroups within the intention-to-treat population, including patients with *EGFR* mutations (both sensitising and non-sensitising; *EGFR*-positive) previously treated with one or more tyrosine kinase inhibitors and patients with baseline liver metastases. Overall survival in the intention-to-treat population was included among secondary efficacy endpoints. Exploratory endpoints included the proportion of patients achieving an objective response in the intention-to-treat population, including *EGFR*-positive patients and patients with baseline liver metastases. Data are reported as per the Jan 22, 2018, data cutoff date, at which the number of coprimary prespecified overall survival events was met in the ABCP versus BCP groups. This trial is registered with ClinicalTrials.gov, number NCT02366143, and is ongoing.

Findings Between March 31, 2015, and Dec 30, 2016, 1202 patients were enrolled. 400 patients were randomly assigned to ABCP, 402 to ACP, and 400 to BCP. In *EGFR*-positive patients (124 of 1202), median overall survival was not estimable (NE; 95% CI 17.0–NE) with ABCP (34 of 400) and 18.7 months (95% CI 13.4–NE) with BCP (45 of 400; hazard ratio [HR] 0.61 [95% CI 0.29–1.28]). Improved overall survival with ABCP versus BCP was observed in patients with sensitising *EGFR* mutations (median overall survival NE [95% CI NE–NE] with ABCP [26 of 400] vs 17.5 months [95% CI 11.7–NE] with BCP [32 of 400]; HR 0.31 [95% CI 0.11–0.83]) and in the intention-to-treat population (19.8 months [17.4–24.2] vs 14.9 months [13.4–17.1]; HR 0.76 [0.63–0.93]). Improved median overall survival with ABCP versus BCP was seen in patients with baseline liver metastases (13.3 months [11.6–NE] with ABCP [52 of 400] vs 9.4 months [7.9–11.7] with BCP [57 of 400]; HR 0.52 [0.33–0.82]). Median overall survival was 21.4 months (95% CI 13.8–NE) with ACP versus 18.7 months (95% CI 13.4–NE) with BCP in *EGFR*-positive patients (HR 0.93 [95% CI 0.51–1.68]). No overall survival benefit was seen with ACP versus BCP in patients with sensitising *EGFR* mutations (HR 0.90 [95% CI 0.47–1.74]), in the intention-to-treat population (HR 0.85 [0.71–1.03]), or in patients with baseline liver metastases (HR 0.87 [0.57–1.32]). In the intention-to-treat safety-evaluable population, grade 3–4 treatment-related events occurred in 223 (57%) patients in the ABCP group, in 172 (43%) in the ACP group, and in 191 (49%) in the BCP group; 11 (3%) grade 5 adverse events occurred in the ABCP group, as did four (1%) in the ACP group, and nine (2%) in the BCP group.

Interpretation Improved survival was noted for patients treated with ABCP compared with those given BCP in the intention-to-treat population, and in patients with baseline liver metastases. The overall survival signal in the subgroup of patients with *EGFR* sensitising mutations warrants further study.

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See Online for appendix

Research in context

Evidence before this study

Treatment of metastatic non-squamous non-small-cell lung cancer in the first-line setting for patients without high programmed death-ligand 1 (PD-L1) expression or oncogenic alterations includes platinum-doublet chemotherapy with or without bevacizumab (anti-vascular endothelial growth factor) or with or without pembrolizumab (anti-programmed death-1 [PD-1]). We searched PubMed for articles and abstracts published up to May, 2018, using the terms “non-small-cell lung cancer,” “PD-L1,” “PD-1,” “chemotherapy,” “bevacizumab,” and “cancer immunotherapy” (full names and abbreviations), which included clinical, non-clinical, English, and non-English publications. The search revealed the therapeutic benefit of anti-PD-L1 and anti-PD-1 agents in combination with chemotherapy in the first-line setting. However, it also showed that recent randomised phase 3 clinical trials excluded patients with epidermal growth factor receptor (EGFR) or anaplastic lymphoma kinase (ALK) genetic alterations because of the minimal efficacy of checkpoint inhibitor monotherapy as second-line or third-line treatment for these patients. Additionally, checkpoint inhibitor monotherapy has shown minimal benefit in patients with liver metastases—a common metastatic site for lung cancer and a negative prognostic indicator. Lastly, bevacizumab, which is approved for first-line treatment of patients with non-squamous non-small-cell lung cancer in combination with carboplatin plus paclitaxel chemotherapy, has shown promising efficacy in patients with liver metastases and in patients with EGFR mutations when combined with the tyrosine kinase inhibitor erlotinib.

Added value of this study

The IMpower150 study was, to the best of our knowledge, the first randomised, open-label, phase 3 trial to report the combination of a checkpoint inhibitor, atezolizumab (anti-PD-L1), with bevacizumab and platinum-doublet chemotherapy in chemotherapy-naïve patients with metastatic non-squamous non-small-cell lung cancer.

The study design also allowed for enrolment of patients with EGFR or ALK genetic alterations and stratified patients by the presence or absence of baseline liver metastases. This approach allowed for the testing of this treatment combination in subgroups of patients who have shown minimal benefit with checkpoint inhibitor therapy and have not been recruited or analysed in other randomised phase 3 studies of checkpoint inhibitor combinations in the first-line setting. This study showed that, in patients with sensitising EGFR mutations (eg, exon 19 deletion or Leu858Arg) and patients with baseline liver metastases, the atezolizumab plus bevacizumab plus carboplatin plus paclitaxel (ABCP) regimen provides an overall and progression-free survival benefit when compared with the standard-of-care bevacizumab plus carboplatin plus paclitaxel (BCP) regimen. Higher proportions of EGFR-positive patients and those with baseline liver metastases achieved an objective response with ABCP than with BCP. By contrast, the atezolizumab plus carboplatin plus paclitaxel (ACP) regimen did not show improved efficacy or survival compared with BCP in these subgroups. Improved overall survival and progression-free survival with ABCP versus BCP was also observed in the intention-to-treat population.

Implications of all the available evidence

The data reported here add to the existing body of evidence that supports the ABCP combination for treatment of chemotherapy-naïve patients with metastatic non-squamous non-small-cell lung cancer. In particular, this report indicates that addition of bevacizumab to atezolizumab and chemotherapy confers the clinical efficacy observed in the two patient subgroups: patients with sensitising EGFR mutations and those with baseline liver metastases. These patient subgroups have shown minimal benefit with checkpoint inhibitor monotherapy in the second-line or later setting, and their needs have not been adequately addressed by currently available regimens following failure of tyrosine kinase inhibitor therapy.

Introduction

Recent advances have expanded treatment options for patients with front-line metastatic non-squamous non-small-cell lung cancer. These options include the use of platinum-doublet chemotherapy with bevacizumab,¹ the use of platinum-pemetrexed with or without pembrolizumab (anti-programmed death-1 [PD-1]),^{2–4} pembrolizumab monotherapy in patients with programmed death-ligand 1 (PD-L1) expression on 50% or more tumour cells,⁵ or tyrosine kinase inhibitors (TKIs) in patients with oncogenic driver mutations or translocations. However, improvements in treatment regimens are still needed. In particular, the use of immune checkpoint inhibitors (anti-PD-1 and anti-PD-L1) as monotherapies have shown minimal therapeutic benefit in some populations, including patients with epidermal

growth factor receptor (EGFR) mutations or baseline liver metastases.

First-line treatment of patients with EGFR mutations by use of TKIs, such as erlotinib, gefitinib, afatinib, and osimertinib, has resulted in median progression-free survival of up to 19 months with osimertinib and around 8–11 months with erlotinib, gefitinib, or afatinib.^{6–12} However, after failure of first-line TKI therapy, progression-free survival with second or subsequent lines of therapy is substantially lower: median progression-free survival of 10·1 months has been reported with osimertinib in the Thr790Met-positive population¹³ or 5·4 months with platinum-doublet chemotherapy with or without gefitinib.¹⁴ Thus, continuation of TKI therapy in combination with chemotherapy is generally associated with worse survival outcomes than with the use of later-generation TKIs, such

as osimertinib, that target specific mutations.^{13,14} Even with Thr790Met resistance mutations (in approximately 60% of patients),¹⁵ most patients will eventually progress and receive chemotherapy.⁶ After unsuccessful first-line chemotherapy, patients with *EGFR* mutations treated with single-agent PD-L1 or PD-1 inhibitors after TKI failure have not shown more substantial survival benefits than patients treated with standard chemotherapy (eg, docetaxel).^{16–18} Therefore, patients with non-small-cell lung cancer who have *EGFR* mutations need better therapies following TKI treatment failure.

Metastases to the liver are common in patients with lung cancer, and these patients have been shown to have a poorer prognosis than patients with metastases to other sites.¹⁹ Furthermore, patients with baseline liver metastases have been shown to have minimal therapeutic benefit with anti-PD-L1 or anti-PD-1 monotherapies.^{20–22} Other chemotherapy combination regimens with nab-paclitaxel or pemetrexed have also shown minimal efficacy in this patient population.^{23,24} Therefore, additional treatment options are needed for these patients.

The anti-PD-L1 antibody atezolizumab inhibits the binding of PD-L1 to its receptors PD-1 and B7.1 to restore antitumour immunity and has shown an overall survival benefit versus docetaxel in patients with previously treated non-small-cell lung cancer regardless of PD-L1 expression. Several ongoing studies in patients with non-small-cell lung cancer are investigating atezolizumab in the first-line setting, including combinations with standard-of-care regimens such as carboplatin plus paclitaxel and bevacizumab.¹ Bevacizumab, in addition to its known anti-angiogenic effects, has immunomodulatory effects through inhibition of vascular endothelial growth factor (VEGF), including promotion of dendritic cell maturation; normalisation of the tumour vasculature, which might increase T-cell infiltration; and reprogramming of the tumour microenvironment from being immune suppressive to immune permissive.^{25,26} Therefore, reversal of VEGF-mediated immunosuppression by bevacizumab could enhance the antitumour activity of atezolizumab.

The randomised phase 3 IMpower150 study aimed to test the efficacy of atezolizumab in combination with bevacizumab plus carboplatin plus paclitaxel (ABCP) or atezolizumab plus carboplatin plus paclitaxel (ACP) versus the standard-of-care bevacizumab plus carboplatin plus paclitaxel (BCP) for the treatment of chemotherapy-naïve patients with metastatic non-squamous non-small-cell lung cancer.²⁷ Significant improvements in progression-free survival and overall survival with ABCP versus BCP were observed in the intention-to-treat wild-type population, and the safety profile of the ABCP combination was shown to be consistent with the safety profiles of the individual drugs.^{1,18,27}

The intention-to-treat population, which included patients with *EGFR* or anaplastic lymphoma kinase (*ALK*) genetic alterations who had progression with, or showed

intolerance to, at least one approved TKI therapy, was a prespecified key secondary population.²⁷ In this report we explored the clinical efficacy of ABCP versus BCP and ACP versus BCP in key subgroups of patients, including patients with *EGFR* mutations and patients with baseline liver metastases.

Methods

Study design and participants

IMpower150 was an international, open-label, randomised, phase 3 study done at 240 academic medical centres and community oncology practices across 26 countries worldwide.²⁷ This study was done in accordance with Good Clinical Practice guidelines and the Declaration of Helsinki, and the study protocol (provided in the appendix) was approved by independent ethics committees at each participating site.²⁷

Eligible patients had chemotherapy-naïve, stage IV metastatic non-squamous non-small-cell lung cancer; measurable disease at baseline according to the Response Evaluation Criteria in Solid Tumours version 1.1 (RECIST v1.1); a baseline Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1; and available tumour tissue for biomarker testing.²⁷ Enrolled patients could have any PD-L1 immunohistochemistry (IHC) status and were required to be eligible to receive bevacizumab.²⁷ Patients with sensitising *EGFR* mutations (exon 19 deletion and Leu858Arg mutations) or *ALK* translocations (tested locally or centrally, or both) must have had disease progression or intolerance to treatment with one or more approved TKI therapy.²⁷ 13 (14%) of 91 patients with sensitising *EGFR* mutations did not receive an approved TKI therapy before randomisation; this was noted within the study accordingly. Additional eligibility criteria for enrolment have been described previously.²⁷ All patients gave written informed consent.²⁷

Randomisation and masking

Patients were stratified according to sex (male vs female), presence of liver metastases at baseline (yes vs no), and PD-L1 expression on tumour cells and tumour-infiltrating immune cells as measured by the SP142 IHC assay (Ventana Medical Systems, Tucson, AZ, USA; TC3 and any tumour-infiltrating immune cell vs TC0/1/2 and IC2/3 vs TC0/1/2 and IC0/1; see appendix for PD-L1 expression subgroup definitions).^{27,28} The presence of baseline liver metastases, which was ascertained by CT scans with contrast imaging or MRI (non-contrast CT was allowed for patients with contraindications, such as an allergy or renal insufficiency), was used as a stratification factor. Permuted block randomisation (block size of six) via an interactive voice or web response system was used to assign patients (1:1:1) to receive ABCP, ACP, or BCP.²⁷ The study was open label, and the study sponsor was masked to treatment effect by population until readout of the co-primary endpoints. Patients were enrolled at the trial centres.

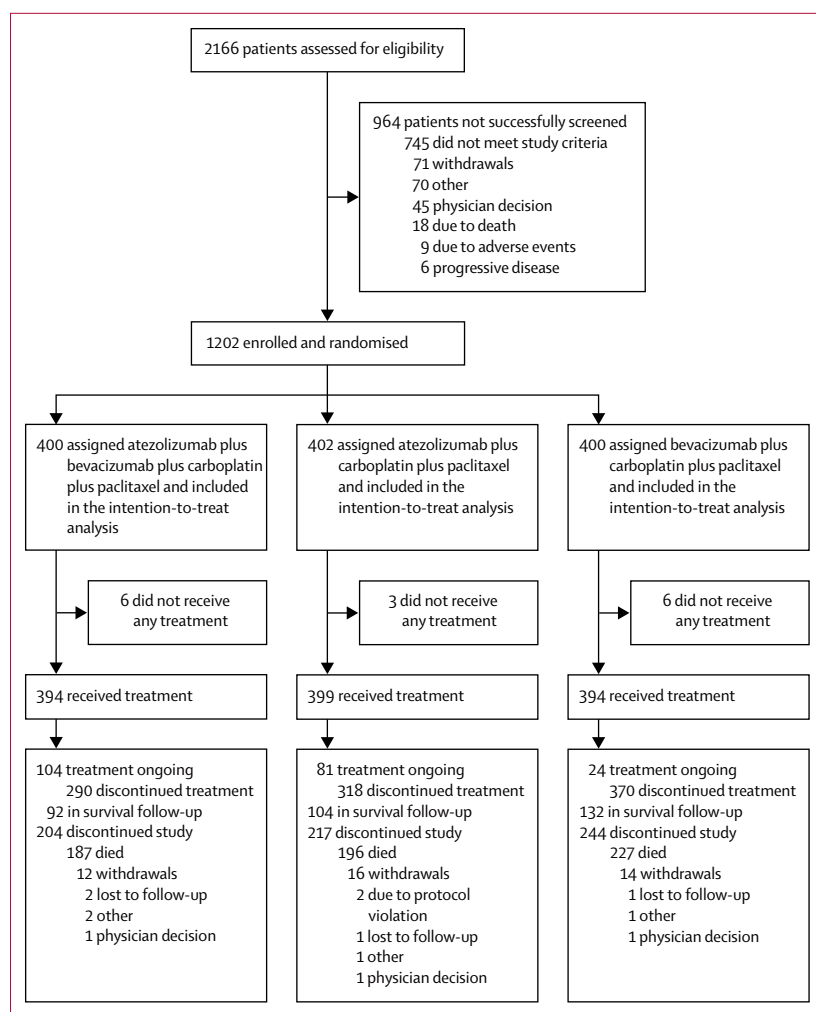


Figure 1: Trial profile

Patients were randomly assigned to receive atezolizumab plus bevacizumab plus carboplatin plus paclitaxel (ABCP), atezolizumab plus carboplatin plus paclitaxel (ACP), or bevacizumab plus carboplatin plus paclitaxel (BCP) every 3 weeks for four or six cycles, followed by maintenance therapy with atezolizumab, bevacizumab, or both. All randomly assigned patients, including those with sensitising and non-sensitising epidermal growth factor receptor (*EGFR*) mutations, comprise the intention-to-treat population. *EGFR*-positive patients were required to have received tyrosine kinase inhibitor therapy as per protocol.

Procedures

Induction chemotherapy treatment was administered for four or six cycles every 21 days as ascertained by the investigator before randomisation.²⁷ Treatment was administered intravenously on day 1 of each 21-day cycle at the following doses: atezolizumab, 1200 mg; bevacizumab, 15 mg/kg; carboplatin, area under the concentration–time curve of 6 mg/mL per min; and paclitaxel, 200 mg/m² (175 mg/m² for patients of Asian ethnicity).²⁷ Following the induction phase, patients receiving ABCP or BCP continued bevacizumab until unmanageable toxicity or disease progression as per RECIST v1.1 as assessed by investigators, and patients receiving ABCP or ACP continued atezolizumab until loss of clinical benefit (including treatment beyond progressive disease as per RECIST v1.1

for patients who met these criteria as ascertained by the investigator).²⁷ As described in the study protocol, because of the potential for pseudoprogression or tumour-immune infiltration, this study allowed patients randomly assigned to atezolizumab treatment to remain on atezolizumab after apparent radiographic progression, provided the benefit–risk ratio was judged to be favourable. Patients were discontinued for unacceptable toxicity, or symptomatic deterioration attributed to disease progression, as determined by the investigator after an integrated assessment of radiographic data and clinical status. No crossover to atezolizumab treatment was permitted.²⁷ However, patients could receive subsequent treatments, including immunotherapy, after withdrawing from study treatment. PD-L1 expression was prospectively analysed in archival or freshly collected tumour tissue, or both, and was centrally analysed with the SP142 PD-L1 IHC assay as described previously.²⁸ Tumour assessments occurred during screening, every 6 weeks from day 1, cycle 1, for the first 48 weeks, and every 9 weeks thereafter until disease progression as per RECIST v1.1 or loss of clinical benefit as applicable for patients who continued atezolizumab treatment beyond initial disease progression.²⁷

Outcomes

The coprimary endpoints of progression-free survival and interim overall survival in the intention-to-treat wild-type population (excluding patients with *EGFR* or *ALK* genetic alterations), and safety data, were previously reported for the ABCP and BCP groups.²⁷ Here, we present key secondary and exploratory efficacy endpoints across all three treatment groups in the following patient populations: patients with *EGFR* mutations, all patients, patients with liver metastases at baseline in the intention-to-treat population, and PD-L1 expression subgroups in the intention-to-treat population. Secondary efficacy endpoints included overall survival and investigator-assessed progression-free survival as per RECIST v1.1 in the intention-to-treat population. The investigator-assessed proportion of patients achieving an objective response and duration of response as per RECIST v1.1 in the intention-to-treat population were exploratory endpoints. Additional exploratory endpoints included overall survival, investigator-assessed progression-free survival, investigator-assessed proportion of patients achieving an objective response, and duration of response in the *EGFR*-positive subgroup and in patients with baseline liver metastases.

Adverse events in the safety-evaluable intention-to-treat and *EGFR*-positive populations are also reported. The frequency, nature, and severity of adverse events was assessed with the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0.

Statistical analysis

The intention-to-treat population in IMpower150 comprised all randomly assigned patients, including

those with *EGFR* or *ALK* genetic alterations. The statistical analysis plan required hierarchical testing of study endpoints in the intention-to-treat wild-type population first, which excluded patients with *EGFR* or *ALK* genetic alterations, and then in the intention-to-treat population.²⁷ The sample size was based on the number of events needed to demonstrate efficacy for the coprimary endpoints of progression-free survival and overall survival. As described in the statistical testing plan for IMpower150,²⁷ a two-sided significance level of 0·05 was used to control for the overall type 1 error rate for the planned primary analyses (progression-free survival in the intention-to-treat wild-type population and wild-type patients with high expression of a T-effector gene signature, and overall survival in the intention-to-treat wild-type population). A brief description of the statistical analyses for the primary endpoints is included in the appendix. The key secondary and exploratory efficacy endpoints in the subgroup analyses described here were not tested under the prespecified type 1 error.

Treatment comparisons for progression-free survival and overall survival in the intention-to-treat population were based on a stratified log-rank test.²⁷ Hazard ratios (HRs) were estimated with a stratified Cox regression model, and 95% CIs were calculated with the Brookmeyer-Crowley method. Stratification factors were those used during randomisation (sex [male vs female], presence of baseline liver metastases [yes vs no], and PD-L1 expression on tumour cells and tumour-infiltrating immune cells as measured by the SP142 IHC assay [TC3 and any IC vs TC0/1/2 and IC2/3 vs TC0/1/2 and IC0/1]). The Kaplan-Meier method was used to estimate medians for progression-free survival, overall survival, and duration of response.²⁷ Subgroup analyses were done with unstratified HRs estimated from a Cox proportional hazards model.²⁷ The proportion of patients achieving an objective response and the corresponding 95% CIs for each treatment group were calculated with the Clopper-Pearson method and compared between treatment groups with the Cochran-Mantel-Haenszel test. Analyses were done with SAS version 9.4, R version 3.3.1, and Spotfire version 7.7. IMpower150 is registered with ClinicalTrials.gov, NCT02366143.

Role of the funding source

F. Hoffmann-La Roche and Genentech funded the study; provided the study drugs; were involved in the study design, data collection, data analysis, data interpretation, and writing of the report; and gave approval to submit for publication. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

Between March 31, 2015, and Dec 30, 2016, 1202 patients across 26 countries were enrolled; 400 patients were

	ABCP group (n=400)	ACP group (n=402)	BCP group (n=400)
Age, years			
Median	63	63	63
Range	31–89	32–85	31–90
Sex			
Male	240 (60%)	241 (60%)	239 (60%)
Female	160 (40%)	161 (40%)	161 (40%)
Race			
White	322 (81%)	331 (82%)	335 (84%)
Asian	56 (14%)	48 (12%)	46 (12%)
Black	3 (1%)	9 (2%)	12 (3%)
American Indian or Alaska Native	3 (1%)	0	1 (<1%)
Multiple	3 (1%)	4 (1%)	0
Unknown	13 (3%)	10 (3%)	6 (1%)
ECOG performance status			
0	159 (40%)	180 (45%)	179 (45%)
1	238 (60%)	222 (55%)	218 (55%)
Smoking history			
Never	82 (21%)	77 (19%)	77 (19%)
Current or previous smoker	318 (80%)	325 (81%)	323 (81%)
Liver metastases at enrolment			
Absent	348 (87%)	349 (87%)	343 (86%)
Present	52 (13%)	53 (13%)	57 (14%)
EGFR mutation status			
Positive	34 (9%)	45 (11%)	45 (11%)
Sensitising mutations*	26 (77%)	33 (73%)	32 (71%)
Negative	353 (88%)	348 (87%)	345 (86%)
Unknown	13 (3%)	9 (2%)	10 (3%)
EML4-ALK rearrangement status			
Positive	11 (3%)	9 (2%)	20 (5%)
Negative	386 (96%)	388 (97%)	376 (94%)
Unknown	3 (1%)	5 (1%)	4 (1%)
PD-L1 subgroup†			
TC3 or IC3	75 (19%)	68 (17%)	73 (18%)
TC1/2 or IC1/2	135 (34%)	148 (37%)	127 (32%)
TC0 and IC0	190 (48%)	185 (46%)	200 (50%)

Data are n (%), unless otherwise indicated. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. ALK=anaplastic lymphoma kinase. BCP=bevacizumab plus carboplatin plus paclitaxel. ECOG=Eastern Cooperative Oncology Group. EGFR=epidermal growth factor receptor. EML4=echinoderm microtubule-associated protein-like 4. PD-L1=programmed death-ligand 1. TC3 refers to PD-L1 expression on 50% or more tumour cells, and IC3 refers to 10% or more tumour-infiltrating immune cells. TC1/2 refers to PD-L1 expression on 1% or more and less than 50% of tumour cells and IC1/2 refers to 1% or more and less than 10% of tumour-infiltrating immune cells. TC0 and IC0 refer to PD-L1 expression on less than 1% of tumour cells and tumour-infiltrating immune cells. *Includes exon 19 deletion and Leu858Arg mutations. †One patient in the ACP group was not tested for PD-L1 expression.

Table 1: Baseline characteristics of the intention-to-treat population

randomly assigned to the ABCP group, 402 to the ACP group, and 400 to the BCP group (figure 1). At the time of data cutoff (Jan 22, 2018), the minimum duration of follow-up was 13·5 months (median follow-up 19·6 months [range 0·0–30·4] for ABCP, 19·6 months

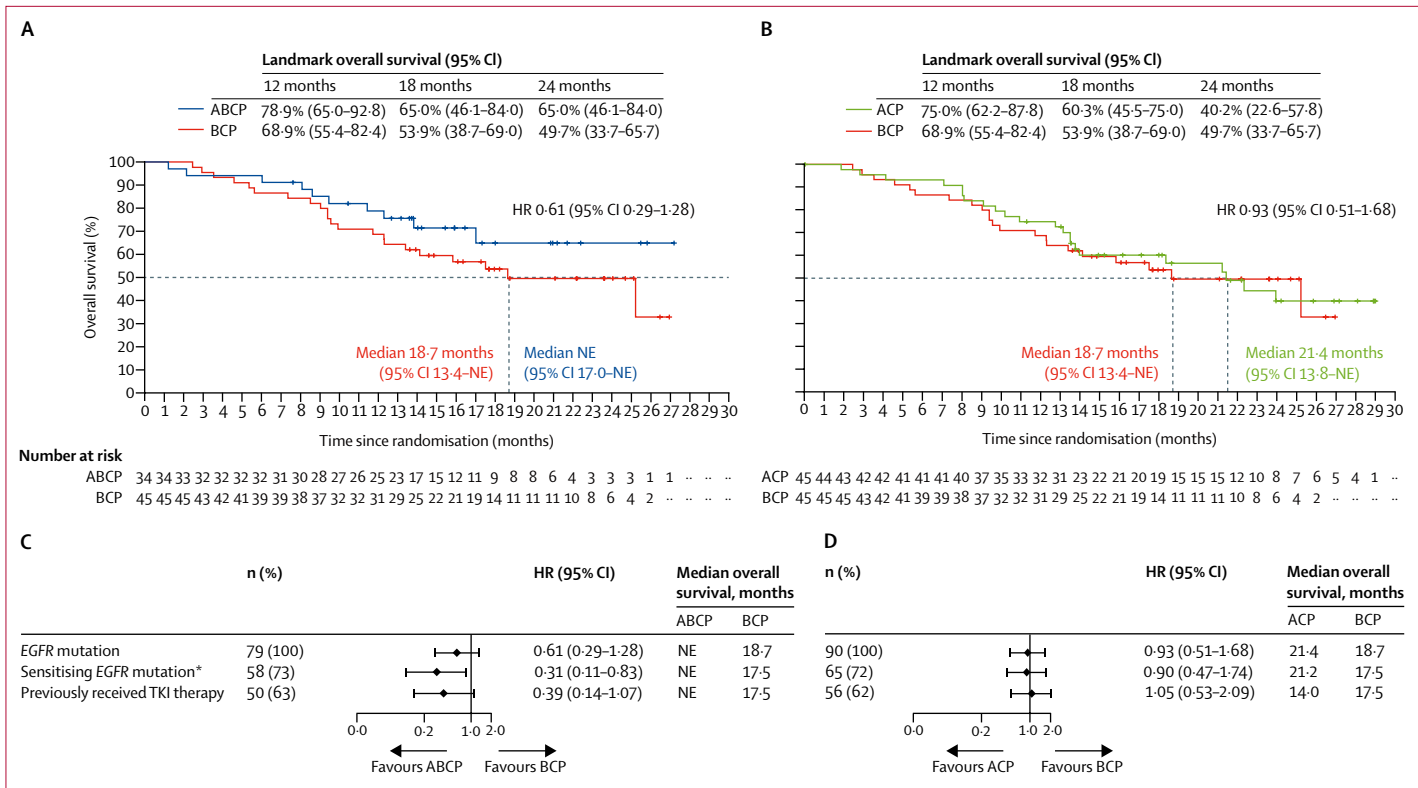


Figure 2: Overall survival in patients with EGFR mutations
Kaplan-Meier estimates in patients with EGFR mutations in the intention-to-treat population for the ABCP versus BCP treatment comparison (A) and the ACP versus BCP treatment comparison (B). HRs for overall survival in patients with EGFR mutations and for those with sensitising EGFR mutations who previously received TKI therapy for the ABCP versus BCP treatment comparison (C) and the ACP versus BCP treatment comparison (D). Median overall survival was estimated by Kaplan-Meier analysis. HRs are unstratified. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. EGFR=epidermal growth factor receptor. HR=hazard ratio. NE=not estimable. TKI=tyrosine kinase inhibitor. *Sensitising EGFR mutations are defined as exon 19 deletions or Leu858Arg mutations. Data cutoff: Jan 22, 2018.

[0.0–29.0] for ACP, and 19.7 months [0.0–32.6] for BCP). Baseline characteristics were well balanced across all treatment groups in the intention-to-treat population (table 1). Approximately 80% of patients in each treatment group were current or previous smokers (table 1). 34 (9%) of 400 patients in the ABCP group were EGFR-positive, as were 45 (11%) of 402 in the ACP group, and 45 (11%) of 400 in the BCP group. Presence of liver metastases at baseline, which was a stratification factor at randomisation, was reported in 52 (13%) of 400 patients in the ABCP group, in 53 (13%) of 402 in the ACP group, and in 57 (14%) of 400 in the BCP group (table 1).

The baseline characteristics of EGFR-positive patients were generally similar to those of the intention-to-treat population (appendix). However, among EGFR-positive patients, four (12%) of 34 patients in the ABCP group, nine (20%) of 45 in the ACP group, and seven (16%) of 45 in the BCP group had baseline liver metastases; and 18 (53%) of 34 patients in the ABCP group, 20 (44%) of 45 in the ACP group, and 27 (60%) of 45 in the BCP group had an ECOG performance status of 0. In the ABCP group, 21 (62%) of 34 EGFR-positive patients had PD-L1-negative tumours (TC0 and IC0; PD-L1 expression on <1% of

tumour cells and tumour-infiltrating immune cells) versus 167 (47%) of 359 patients in the intention-to-treat wild-type population (appendix). Among EGFR-positive patients, four (12%) of 34 in the ABCP group, four (9%) of 45 in the ACP group, and five (11%) of 45 in the BCP group had tumours with high PD-L1 expression (TC3 or IC3; PD-L1 expression on ≥50% of tumour cells or ≥10% of tumour-infiltrating immune cells) compared with 71 (20%) of 359 in the ABCP group, 63 (18%) of 349 in the ACP group, and 65 (19%) of 337 in the BCP group, among patients in the intention-to-treat wild-type population.

An exon 19 deletion and Leu858Arg mutation were the most common EGFR mutation types; three patients had a Thr790Met mutation (appendix). For analyses of patients who previously received TKI therapy, only those with an exon 19 deletion or Leu858Arg mutation were considered. Of these patients, 22 (85%) of 26 in the ABCP group, 28 (85%) of 33 in the ACP group, and 28 (88%) of 32 in the BCP group previously received at least one TKI therapy. Among patients with an exon 19 deletion or Leu858Arg mutation, previous TKIs received included erlotinib (11 [42%] of 26 patients in the ABCP group, 13 [39%] of 33 in the ACP group, and

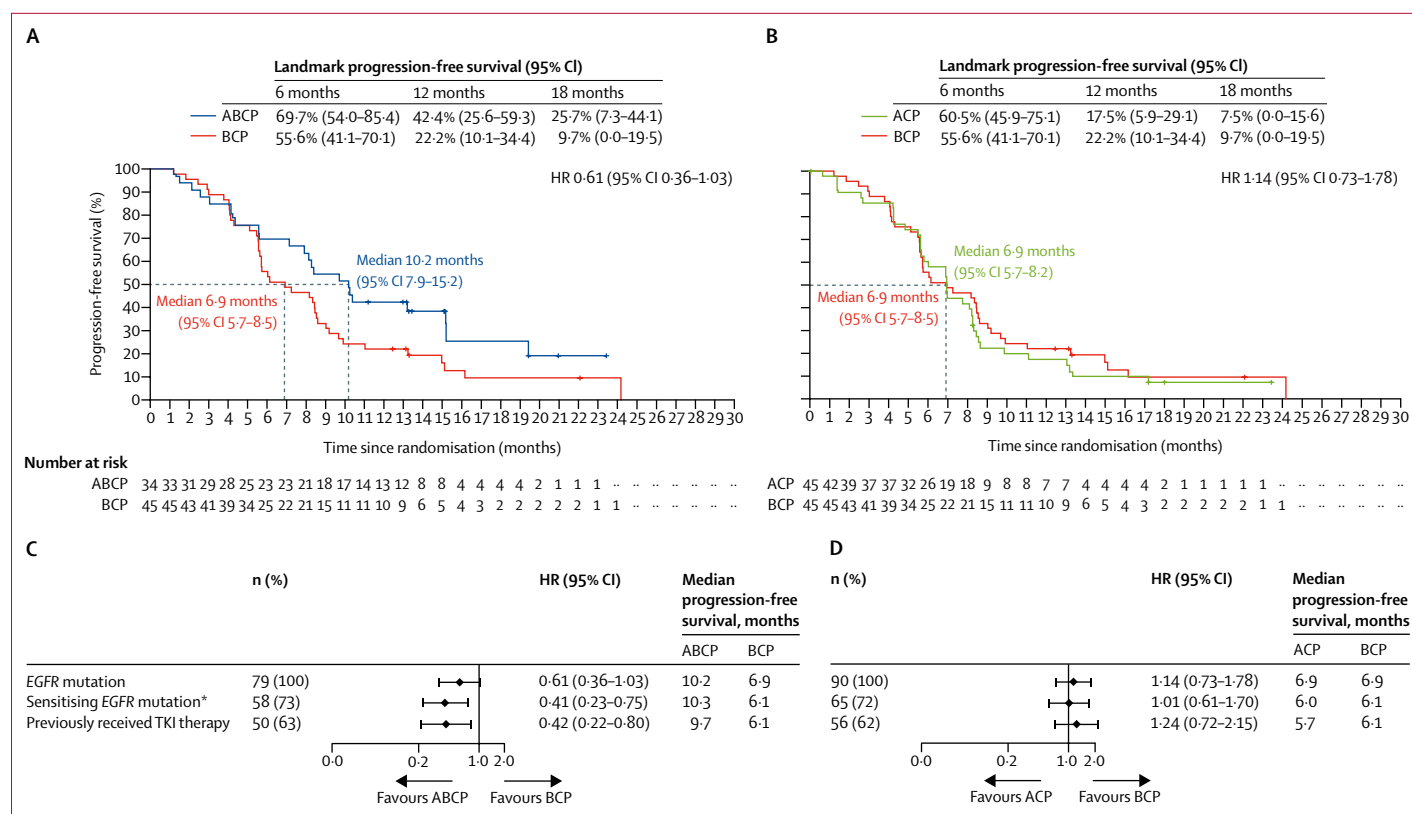


Figure 3: Investigator-assessed progression-free survival in patients with EGFR mutations

Kaplan-Meier estimates in patients with EGFR mutations in the intention-to-treat population for the ABCP versus BCP treatment comparison (A) and the ACP versus BCP treatment comparison (B). HRs for progression-free survival in patients with EGFR mutations and for those with sensitising EGFR mutations who previously received TKI therapy for the ABCP versus BCP treatment comparison (C) and the ACP versus BCP treatment comparison (D). Median progression-free survival was estimated by Kaplan-Meier analysis. HRs are unstratified. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. HR=hazard ratio. TKI=tyrosine kinase inhibitor. *Sensitising EGFR mutations are defined as exon 19 deletions or Leu858Arg mutations. Data cutoff: Jan 22, 2018.

12 [38%] of 32 in the BCP group); gefitinib (nine [35%] of 26 patients in the ABCP group, seven [21%] of 33 in the ACP group, and 11 [34%] of 32 in the BCP group); and afatinib (four [15%] of 26 patients in the ABCP group, eight [24%] of 33 in the ACP group, and seven [22%] of 32 in the BCP group). One patient in the ABCP group, four in the ACP group, and five in the BCP group previously received osimertinib.

The overall survival HR for ABCP versus BCP in all EGFR-positive patients was 0.61 (95% CI 0.29–1.28; figure 2A). Despite the HR point estimate of 0.61, the upper bound 95% CI crossed 1, potentially because of the small sample size and insufficient power. Median overall survival was not estimable (NE) in 34 of 400 patients (95% CI 17.0–NE) for ABCP and 18.7 months (95% CI 13.4–NE) in 45 of 400 patients for BCP. In the subgroup of patients with sensitising EGFR mutations, the ABCP regimen showed an improvement in overall survival compared with the BCP regimen; median overall survival with ABCP was NE (95% CI NE–NE; 26 of 400 patients) and 17.5 months (95% CI 11.7–NE; 32 of 400 patients) with BCP (HR 0.31 [95% CI 0.11–0.83]; figure 2C; appendix). In patients with sensitising EGFR mutations

who previously received EGFR TKI therapy, the HR for overall survival was 0.39 (95% CI 0.14–1.07; figure 2C; appendix). The HR for progression-free survival with ABCP versus BCP in EGFR-positive patients was 0.61 (95% CI 0.36–1.03), with a median progression-free survival of 10.2 months (95% CI 7.9–15.2) versus 6.9 months (5.7–8.5; figure 3A). Patients with sensitising EGFR mutations showed improved progression-free survival with ABCP versus BCP (HR 0.41 [95% CI 0.23–0.75]; figure 3C). 24 (71%) of 34 EGFR-positive patients achieved an objective response with ABCP versus 18 (42%) of 43 with BCP (table 2). The median duration of response was 11.1 months (range 2.8–18.0) with ABCP and 4.7 months (range 2.6–13.5) with BCP; nine (38%) of 24 patients with ABCP and none with BCP had an ongoing response at the time of data cutoff. 21 (81%) of 26 (95% CI 60.7–93.5) patients with sensitising EGFR mutations achieved an objective response with ABCP and the median duration of response was 11.1 months (range 2.8–18.0); eight of 21 patients had an ongoing response at the time of data cutoff. In the ABCP group, a higher proportion of patients with EGFR mutations achieved an objective

	ABCP group	ACP group	BCP group
Intention-to-treat population†			
Number of patients	397	401	393
Proportion of patients with an objective response	224 (56.4%; 95% CI 51.4–61.4)	163 (40.6%; 95% CI 35.8–45.6)	158 (40.2%; 95% CI 35.3–45.2)
Complete response	11 (2.8%; 95% CI 1.4–4.9)	8 (2.0%; 95% CI 0.9–3.9)	3 (0.8%; 95% CI 0.2–2.2)
Partial response	213 (53.7%; 95% CI 48.6–58.6)	155 (38.7%; 95% CI 33.9–43.6)	155 (39.4%; 95% CI 34.6–44.5)
Stable disease	111 (28.0%; 95% CI 23.6–32.7)	154 (38.4%; 95% CI 33.6–43.4)	160 (40.7%; 95% CI 35.8–45.8)
Progressive disease	23 (5.8%; 95% CI 3.7–8.6)	58 (14.5%; 95% CI 11.2–18.3)	38 (9.7%; 95% CI 6.9–13.0)
Median duration of response, months (range)	11.5 (2.0–29.0)	8.3 (1.9–26.0‡)	6.0 (1.5‡–23.1‡)
Number of patients with ongoing response at cutoff	88 (39.3%)	56 (34.4%)	18 (11.4%)
EGFR-positive mutation§¶			
Number of patients	34	45	43
Proportion of patients with an objective response	24 (70.6%; 95% CI 52.5–84.9)	16 (35.6%; 95% CI 21.9–51.2)	18 (41.9%; 95% CI 27.0–57.9)
Complete response	2 (5.9%; 95% CI 0.7–19.7)	1 (2.2%; 95% CI 0.1–11.8)	0
Partial response	22 (64.7%; 95% CI 46.5–80.3)	15 (33.3%; 95% CI 20.0–49.0)	18 (41.9%; 95% CI 27.0–57.9)
Stable disease	5 (14.7%; 95% CI 5.0–31.1)	21 (46.7%; 95% CI 31.7–62.1)	19 (44.2%; 95% CI 29.1–60.1)
Progressive disease	2 (5.9%; 95% CI 0.7–19.7)	6 (13.3%; 95% CI 5.1–26.8)	3 (7.0%; 95% CI 1.5–19.1)
Median duration of response, months (range)	11.1 (2.8–18.0)	5.6 (2.6–15.2)	4.7 (2.6–13.5)
Number of patients with ongoing response at cutoff	9 (37.5%)	3 (18.8%)	0
Liver metastases at baseline 			
Number of patients	51	52	56
Proportion of patients with an objective response	31 (60.8%; 95% CI 46.1–74.2)	14 (26.9%; 95% CI 15.6–41.0)	23 (41.1%; 95% CI 28.1–55.0)
Complete response	0	0	0
Partial response	31 (60.8%; 95% CI 46.1–74.2)	14 (26.9%; 95% CI 15.6–41.0)	23 (41.1%; 95% CI 28.1–55.0)
Stable disease	9 (17.6%; 95% CI 8.4–30.9)	21 (40.4%; 95% CI 27.0–54.9)	16 (28.6%; 95% CI 17.3–42.2)
Progressive disease	3 (5.9%; 95% CI 1.2–16.2)	11 (21.2%; 95% CI 11.1–34.7)	10 (17.9%; 95% CI 8.9–30.4)
Median duration of response, months (range)	10.7 (2.8–24.8)	5.6 (2.0–19.0)	4.6 (2.8–22.1)
Number of patients with ongoing response at cutoff	8 (25.8%)	2 (14.0%)	0
Data are n (%) or n (%; 95% CI), unless otherwise indicated. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. EGFR=epidermal growth factor receptor. *Objective response was defined as a confirmed complete response or partial response, as ascertained by the investigator according to Response Evaluation Criteria in Solid Tumours, version 1.1. Only patients with measurable disease at baseline were included in the analysis of patients achieving an objective response. †Missing or unevaluable in the intention-to-treat population: 39 patients in the ABCP group, 26 patients in the ACP group, and 37 patients in the BCP group. ‡Censored value. §One patient in the ABCP group had both an EGFR exon 19 deletion and tested positive for ALK rearrangement by central laboratory. ¶Missing or unevaluable in the EGFR-positive subgroup: three patients in the ABCP group, two patients in the ACP group, and three patients in the BCP group. One patient in the BCP group had a non-complete response or non-progressive disease response. Missing or unevaluable in the subgroup with liver metastases at baseline: nine patients in the ABCP group, six patients in the ACP group, and seven patients in the BCP group. Data cutoff: Jan 22, 2018.			
Table 2: Proportion of patients achieving an objective response* and duration of response in the intention-to-treat population, in patients with EGFR mutations, and in patients with baseline liver metastases			

response compared with the intention-to-treat population (table 2). Results observed in patients with *EGFR* mutations in the ABCP group did not appear to be driven by high PD-L1 expression (appendix).

Median overall survival for *EGFR*-positive patients with ACP versus BCP was 21.4 months (95% CI 13.8–NE) versus 18.7 months (13.4–NE; HR 0.93 [95% CI 0.51–1.68]) and progression-free survival was 6.9 months (95% CI 5.7–8.2) versus 6.9 months (5.7–8.5; HR 1.14 [95% CI 0.73–1.78]; figure 2B, 2D; figure 3B, 3D). 16 (36%) of 45 *EGFR*-positive patients achieved an objective response with ACP, with a median duration of response of 5.6 months (range 2.6–15.2; table 2).

Overall, one (3%) of 34 *EGFR*-positive patients in the ABCP group, three (7%) of 45 in the ACP group, and nine (20%) of 45 in the BCP group, received subsequent

non-protocol immunotherapy, whereas ten (29%) of 34 patients in the ABCP group, 22 (49%) of 45 in the ACP group, and 21 (47%) of 45 in the BCP group received a subsequent targeted therapy. Of those patients who received a subsequent targeted therapy, six (18%) of 34 patients in the ABCP group, seven (16%) of 45 in the ACP group, and ten (22%) of 45 in the BCP group received osimertinib.

In the intention-to-treat population, ABCP showed improved overall survival versus BCP (HR 0.76 [95% CI 0.63–0.93]; median overall survival 19.8 months [95% CI 17.4–24.2] vs 14.9 months [13.4–17.1]; figure 4A). Improved progression-free survival with ABCP versus BCP was also observed (HR 0.59 [95% CI 0.50–0.69]; median progression-free survival 8.4 months [95% CI 8.0–9.9] vs 6.8 months [6.0–7.0]; appendix). 224 (56%)

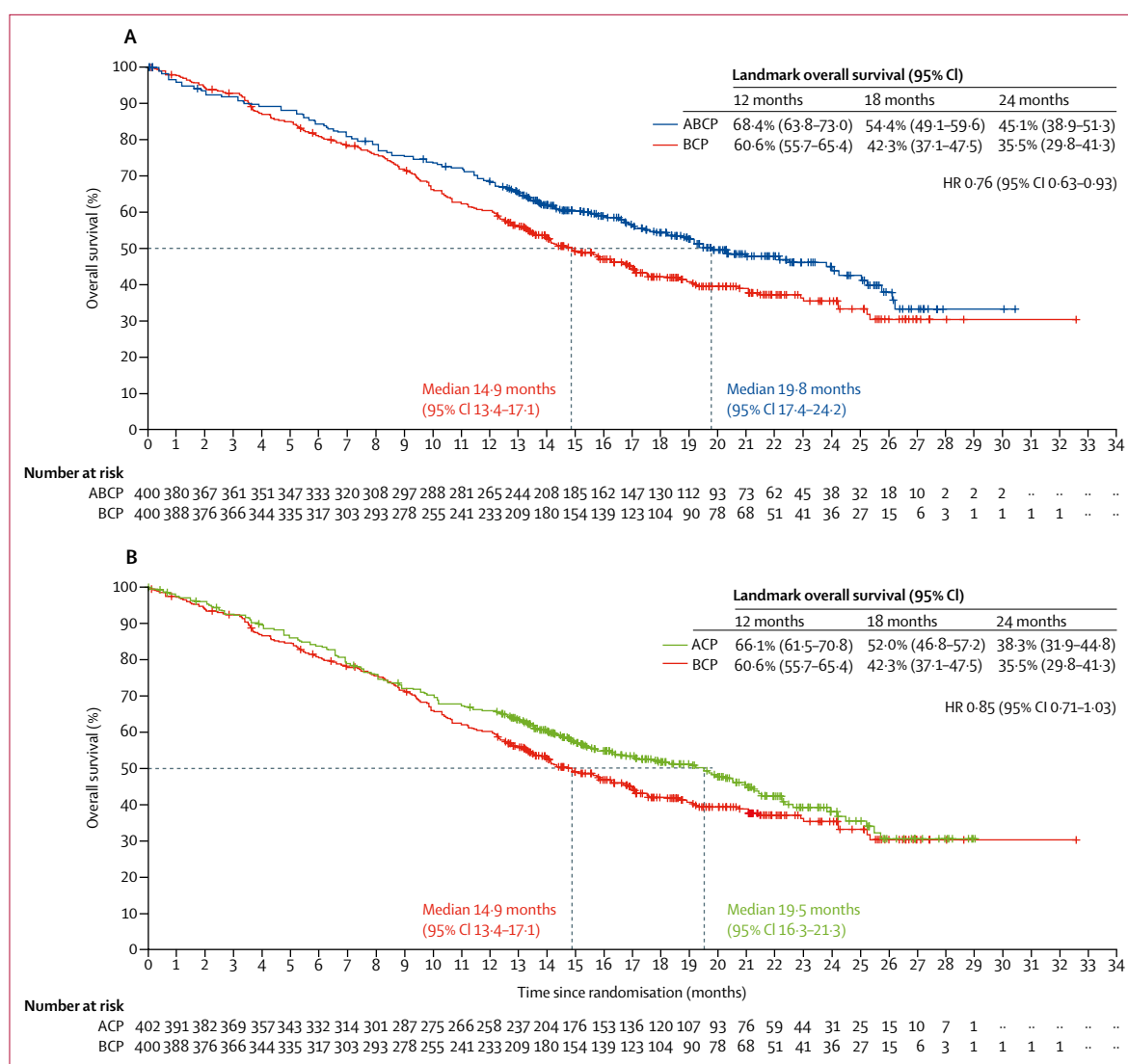


Figure 4: Overall survival in the intention-to-treat population

Kaplan-Meier estimates in the intention-to-treat population for the ABCP versus BCP treatment comparison (A) and the ACP versus BCP treatment comparison (B). Median overall survival was estimated by Kaplan-Meier analysis. HRs are stratified. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. HR=hazard ratio. Data cutoff: Jan 22, 2018.

of 397 patients achieved an objective response with ABCP, as did 158 (40%) of 393 with BCP (table 2). The median duration of response was 11.5 months (range 2.0–29.0) with ABCP and 6.0 months (range 1.5–23.1) with BCP; 88 (39%) of 224 patients with ABCP and 18 (11%) of 158 with BCP had an ongoing response at the time of data cutoff.

Median overall survival in the intention-to-treat population was 19.5 months (95% CI 16.3–21.3) with ACP versus 14.9 months (13.4–17.1) with BCP (HR 0.85 [95% CI 0.71–1.03]; figure 4B). The HR for progression-free survival with ACP versus BCP was 0.91 (95% CI 0.78–1.06; appendix). 163 (41%) of 401 patients achieved an objective response with ACP, with a median duration of response of 8.3 months (range 1.9–26.0; table 2).

An improvement in overall survival favouring ABCP versus BCP was observed across several demographic and baseline characteristic subgroups in the intention-to-treat population, including male patients (HR 0.73 [95% CI 0.57–0.93]) and patients aged 65–74 years (HR 0.69 [0.49–0.96]; appendix). However, improved overall survival with ACP versus BCP was not observed across most baseline characteristic subgroups (appendix).

Efficacy analyses were done in the subgroup of patients in the intention-to-treat population who had liver metastases at baseline. 52 of 400 patients with baseline liver metastases had improved overall survival with ABCP versus 57 of 400 with BCP, with a median overall survival of 13.3 months (95% CI 11.6–NE) with ABCP versus 9.4 months (7.9–11.7) with BCP (HR 0.52 [95% CI

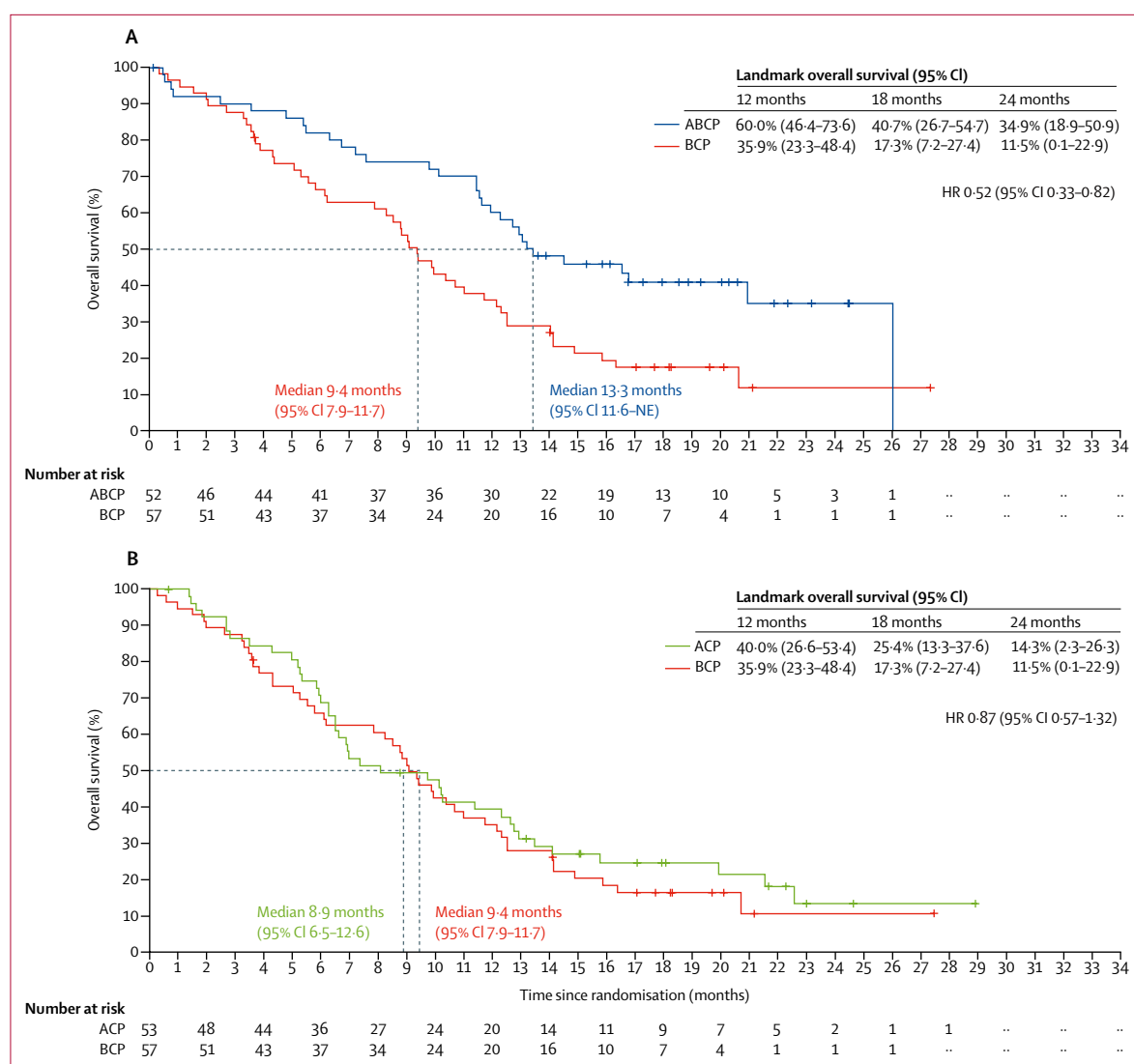


Figure 5: Overall survival in patients with liver metastases at baseline in the intention-to-treat population

Kaplan-Meier estimates in patients with liver metastases at baseline in the intention-to-treat population for the ABCP versus BCP treatment comparison (A) and the ACP versus BCP treatment comparison (B). HRs are unstratified. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. HR=hazard ratio. NE=not estimable. Data cutoff: Jan 22, 2018.

0.33–0.82]; figure 5A). Improved progression-free survival for patients with liver metastases was also observed with ABCP versus BCP (median 8.2 months [95% CI 5.7–10.3] vs 5.4 months [4.1–6.0]; HR 0.41 [95% CI 0.26–0.62]; figure 6A). 31 (61%) of 51 patients achieved an objective response with ABCP compared with 23 (41%) of 56 patients with BCP (table 2). The median duration of response was 10.7 months (range 2.8–24.8) with ABCP and 4.6 months (range 2.8–22.1) with BCP; eight (26%) of 31 patients in the ABCP group and none in the BCP group had an ongoing response at the time of data cutoff (table 2). As was done in *EGFR*-positive patients, we confirmed that the efficacy observed in patients in the ABCP group with baseline liver metastases was not driven by high PD-L1 expression (appendix).

No improvement in overall survival or progression-free survival was observed for ACP versus BCP in patients with baseline liver metastases (figure 5B; figure 6B). 14 (27%) of 52 patients with baseline liver metastases achieved an objective response with ACP, with a median duration of response of 5.6 months (range 2.0–19.0; table 2).

Efficacy analyses were also done across PD-L1 expression subgroups in the intention-to-treat population. In patients with low PD-L1 expression (TC1/2 or IC1/2; PD-L1 expression on $\geq 1\%$ and $< 50\%$ of tumour cells and $\geq 1\%$ and $< 10\%$ of tumour-infiltrating immune cells) or PD-L1-negative tumours, median overall survival was 22.5 months (95% CI 17.0–26.2) with ABCP versus 16.7 months (95% CI 12.5–22.9) with BCP (HR 0.76

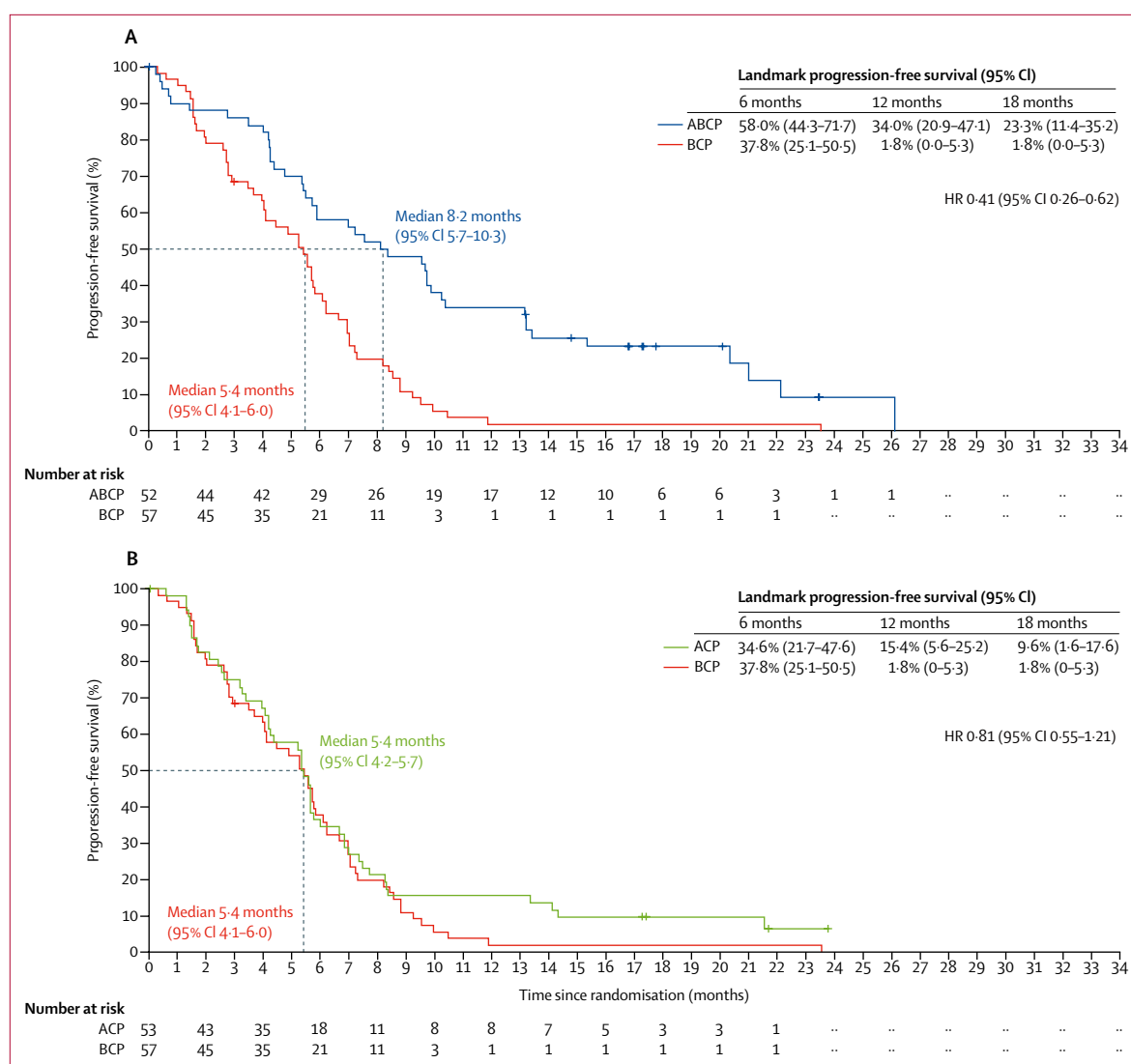


Figure 6: Investigator-assessed progression-free survival in patients with liver metastases at baseline

Kaplan-Meier estimates in patients with liver metastases at baseline in the intention-to-treat population for the ABCP versus BCP treatment comparison (A) and the ACP versus BCP treatment comparison (B). HRs are unstratified. ABCP=atezolizumab plus bevacizumab plus carboplatin plus paclitaxel. ACP=atezolizumab plus carboplatin plus paclitaxel. BCP=bevacizumab plus carboplatin plus paclitaxel. HR=hazard ratio. Data cutoff: Jan 22, 2018.

[95% CI 0.54–1.08]) and 17.1 months (95% CI 13.8–21.0) with ABCP versus 14.4 months (95% CI 13.4–16.9) with BCP (HR 0.83 [95% CI 0.64–1.08]; appendix). In both PD-L1 expression subgroups, improved progression-free survival was observed with ABCP versus BCP (low PD-L1 expression: HR 0.55 [95% CI 0.42–0.73]; PD-L1-negative tumours: HR 0.75 [95% CI 0.60–0.94]; appendix). Median overall survival in patients with PD-L1-high tumours was 25.2 months (95% CI 19.5–NE) with ABCP versus 13.2 months (95% CI 9.8–NE) with BCP (HR 0.67 [95% CI 0.42–1.06]; appendix). High PD-L1 expression contributed to improved progression-free survival with ABCP versus BCP (HR 0.33 [95% CI 0.22–0.51]; appendix). A higher proportion of patients achieved an objective response with ABCP than with BCP across all

PD-L1 expression subgroups, including PD-L1-high (51 [69%] of 74 vs 35 [49%] of 71), PD-L1-low (78 [58%] of 134 vs 51 [41%] of 126), and PD-L1-negative groups (95 [50%] of 189 vs 72 [37%] of 196; appendix). The duration of response was longer for patients in the ABCP group than in the BCP groups across all PD-L1 subgroups, with a higher proportion of patients in the ABCP groups having an ongoing response at the time of data cutoff than those in the BCP group (appendix).

Median overall survival was 24.2 months (95% CI 18.4–25.6) with ACP versus 16.7 months (95% CI 12.5–22.9) with BCP (HR 0.74 [95% CI 0.52–1.03]) in the PD-L1-low subgroup, and 15.0 months (13.1–19.5) with ACP versus 14.4 months (13.4–16.9) with BCP in the PD-L1-negative subgroup (HR 0.98 [0.76–1.27];

appendix). Median progression-free survival was 6·9 months (95% CI 5·8–8·2) with ACP versus 6·2 months (5·7–7·1) with BCP in the PD-L1-low subgroup (HR 0·79 [95% CI 0·61–1·03]) and 5·5 months (4·8–6·7) with ACP versus 6·9 months (5·9–7·8) with BCP in the PD-L1-negative subgroup (HR 1·10 [0·89–1·36]; appendix). In the PD-L1-high subgroup, improved progression-free survival was observed with ACP versus BCP (HR 0·63 [95% CI 0·43–0·92]). The proportion of patients achieving an objective response was similar between the ACP and BCP groups in the PD-L1-low and PD-L1-negative subgroups (appendix). A higher proportion of patients achieved an objective response with ACP than with BCP in the PD-L1-high subgroup (42 [62%] of 68 vs 35 [49%] of 71). Median duration of response was prolonged for ACP versus BCP across all PD-L1 subgroups.

The frequency of all-cause, treatment-related, and serious adverse events in safety-evaluable patients of the intention-to-treat and *EGFR*-positive populations is shown in the appendix. In the safety-evaluable intention-to-treat population, grade 3–4 adverse events were observed in 250 (64%) of 393 patients in the ABCP group, in 230 (58%) of 400 in the ACP group, and in 230 (58%) of 394 in the BCP group. In the *EGFR*-positive safety-evaluable population, grade 3–4 adverse events were observed in 21 (64%) of 33 patients in the ABCP group, 30 (68%) of 44 in the ACP group, and 28 (64%) of 44 in the BCP group. No patients in the ABCP and ACP groups in the *EGFR*-positive population had a grade 5 adverse event; one patient in the BCP group had a grade 5 adverse event. In the intention-to-treat safety-evaluable population, grade 3–4 treatment-related events occurred in 223 (57%) patients in the ABCP group, in 172 (43%) in the ACP group, and in 191 (49%) in the BCP group. Grade 5 treatment-related adverse events among the intention-to-treat population were observed in 11 (3%) of 393 patients with ABCP, in four (1%) of 400 with ACP, and in nine (2%) of 394 with BCP. Adverse events of special interest, which included immune-related adverse events, were observed in 206 (52%) of 393 patients with ABCP, 192 (48%) of 400 with ACP, and 112 (28%) of 394 with BCP among the safety-evaluable intention-to-treat population; and in 18 (55%) of 33 patients with ABCP, in 23 (52%) of 44 with ACP, and in ten (23%) of 44 with BCP among the safety-evaluable *EGFR*-positive population.

Discussion

This study is, to the best of our knowledge, the first randomised phase 3 trial of a checkpoint inhibitor in combination with chemotherapy and anti-VEGF therapy to show an improvement in progression-free survival and overall survival in previously treated patients with sensitising *EGFR* mutations and patients with baseline liver metastases. The overarching hypothesis of this study was based on the potential additive or synergistic effects of combining immunotherapy with chemotherapy-induced

neoantigen release.²⁹ Furthermore, the anti-angiogenic and immunomodulatory effects of bevacizumab, which can lead to immune reprogramming of the tumour microenvironment from an immune-suppressive to an immune-permissive one, make it an attractive combination partner in this regimen.^{25,26} The mechanism of action for the treatment effects in *EGFR*-positive patients and those with baseline liver metastases is still being explored.

The addition of atezolizumab to standard-of-care bevacizumab and carboplatin plus paclitaxel in this phase 3 study of chemotherapy-naïve patients with metastatic non-squamous non-small-cell lung cancer has shown a 24% reduced risk of death, improvement in progression-free survival, and a higher proportion of patients achieving a response than with bevacizumab and chemotherapy in the overall intention-to-treat population, which included *EGFR*-positive patients and those with baseline liver metastases.

The data suggest an improvement in progression-free survival with the combined ABCP regimen in *EGFR*-positive patients compared with the BCP regimen. By contrast, previous data have shown no benefit with anti-PD-L1 and anti-PD-1 monotherapy compared with standard-of-care chemotherapy in these patients in the second-line setting.^{16–18} Furthermore, a lower prevalence of PD-L1 expression was observed in *EGFR*-positive patients than in wild-type patients, suggesting that the observed clinical benefit was not driven by high PD-L1 expression. Reduced CD8+ T-cell tumour infiltration has been shown in *EGFR*/*ALK*-positive patients both before and after TKI therapy.³⁰ Because bevacizumab-induced tumour vasculature normalisation can promote T-cell tumour infiltration,^{25,26} treatment of *EGFR*-positive patients with bevacizumab might increase their sensitivity to checkpoint inhibitor therapy.

The observation of improved progression-free survival with ABCP versus BCP, without a similar benefit with ACP versus BCP, in patients with sensitising *EGFR* mutations suggests that the combination of atezolizumab and bevacizumab added on to the carboplatin plus paclitaxel regimen could provide benefit in this patient population. *EGFR* signalling has been shown to promote VEGF expression in tumours,^{31,32} which might enhance the sensitivity of patients with *EGFR* mutations to bevacizumab. Indeed, bevacizumab was previously shown to provide clinical benefit to patients with sensitising *EGFR* mutations when combined with the *EGFR* TKI erlotinib than with erlotinib alone.^{33,34}

Previous studies of checkpoint inhibitors have shown minimal therapeutic benefit as a single-agent therapy^{20–22} or in combination with chemotherapy (IMpower130: carboplatin plus nab-paclitaxel [NCT02367781]; IMpower132: carboplatin or cisplatin plus pemetrexed [NCT02657434])^{23,24} in patients with baseline liver metastases. In other studies of checkpoint inhibitor-chemotherapy combinations, data in patients with

baseline liver metastases have not been reported.^{2,35} In this study, a benefit with ABCP versus BCP was seen despite lower PD-L1 expression in patients with liver metastases than in patients without liver metastases at baseline, suggesting that the observation was not driven by high PD-L1 expression. Hepatocellular carcinoma is associated with hypoxic tumour conditions, high VEGF expression, and increased angiogenesis, which can contribute to the induction of immunosuppressive immune-cell types (eg, myeloid-derived suppressor cells and regulatory T cells) and the promotion of immune tolerance in the tumour microenvironment.^{36–38} Liver metastases from lung cancer have been shown to respond to treatment in a more similar way to liver cancer than to lung cancer.^{21,39} Furthermore, bevacizumab has been shown to improve overall survival in patients with baseline liver metastases when added to carboplatin plus paclitaxel.¹ Therefore, the poor response of patients with liver metastases to immune checkpoint inhibitor monotherapy might be due to tissue-specific immunoregulation and might be reversed by the addition of bevacizumab. Indeed, improvements in overall and progression-free survival were seen with ABCP in this patient population compared with ACP in this study. The observation that ACP did not prolong overall survival or progression-free survival in *EGFR*-positive patients or in patients with baseline liver metastases compared with BCP further supports the hypothesis that it is the addition of atezolizumab and bevacizumab to the chemotherapy regimen that provides the observed benefit to these patient subgroups.

The randomised phase 3 IMpower130 study of atezolizumab plus carboplatin and nab-paclitaxel versus carboplatin and nab-paclitaxel also enrolled *EGFR*/ALK-positive patients and stratified the intention-to-treat population by the presence of baseline liver metastases. Although this study met its co-primary endpoints of progression-free survival and overall survival in the experimental group of the intention-to-treat wild-type population, outcomes for the *EGFR*/ALK-positive subgroup of patients appeared to be similar between the experimental and control groups, thus supporting the need for bevacizumab in combination with atezolizumab and chemotherapy in these patient subgroups.²³

No improvement in overall survival, progression-free survival, or the proportion of patients achieving an objective response was observed with ACP versus BCP in the intention-to-treat population. Further follow-up of the ACP group is needed, and, as the overall survival efficacy boundary for the ACP versus BCP treatment comparison has not been crossed in the primary study population (intention-to-treat wild-type patients) at the interim overall survival analysis, this comparison will be tested again at the time of the final overall survival analysis.

The inclusion of patients with *EGFR* mutations after unsuccessful TKI therapy allowed for the testing of the

ABCP and ACP experimental treatments in this clinically relevant patient population. Other large phase 3 clinical trials of checkpoint inhibitor combinations in patients with metastatic non-squamous non-small-cell lung cancer treated in the first-line setting excluded patients with *EGFR* genetic alterations.^{2,40} Furthermore, stratification by the presence of liver metastases at baseline ensured a balanced number of patients among the ABCP, ACP, and BCP groups in the intention-to-treat population. However, the small sample size of each subgroup is a limitation of this analysis and has resulted in imbalances in mutation type, smoking history, previous TKI use, and potentially other subgroups. Furthermore, although these analyses were prespecified, they were exploratory in nature and were therefore not sufficiently powered to detect differences between treatment regimens with statistical rigour. Thus, these data must be interpreted with caution. Further analysis of key subgroups in IMpower150 and in other studies, such as IMpower130, will elucidate the role of atezolizumab plus chemotherapy with or without bevacizumab in these key populations. Overall, the findings of these subgroup analyses warrant further investigation. Results from current and future studies assessing the combination of chemotherapy and immunotherapy in *EGFR*-positive patients will provide additional insights.

In summary, the addition of atezolizumab to standard-of-care bevacizumab and carboplatin plus paclitaxel chemotherapy appears to improve survival outcomes in an all-comer population of chemotherapy-naïve patients with metastatic non-squamous non-small-cell lung cancer. This randomised phase 3 study is, to the best of our knowledge, the first to show promising results with the combination of atezolizumab plus bevacizumab and chemotherapy in patients with *EGFR* mutations and patients with baseline liver metastases. The ABCP regimen might represent a potential new therapy option for these key patient subgroups, especially those with sensitising *EGFR* mutations who have progressed on TKI therapy.

Contributors

All authors reviewed the data analyses, contributed to data interpretation and the writing of the manuscript, approved the final version of the submitted manuscript, and agreed to be accountable for all aspects of the report.

Declaration of interests

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Chi-Med, Ignyta, Janssen, Merck Serono, MSD, Novartis, OncoGenex Technologies, OrigiMed, Pfizer, Roche/Genentech, Sanofi-Aventis, SFJ Pharmaceutical, Takeda, Vertex Pharmaceuticals, Clovis Oncology, and Yuhan Corporation; has received speaker fees from AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Eli Lilly, MSD, Novartis, Pfizer, Roche/Genentech, Takeda, Taiho Pharmaceutical, and Amoy Diagnostics; and has stock options in Loxo-Oncology, Hutchison Chi-Med as independent non-executive director, AstraZeneca as non-executive director, and Sanomics as director. MN has received research grants from Astellas, AstraZeneca, Bristol-Myers Squibb, Chugai Pharmaceutical, Eli Lilly, MSD, Novartis, Ono Pharmaceutical, Pfizer, and Taiho Pharmaceutical, and has participated in speakers' bureaus or advisory boards for AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Chugai Pharmaceutical, Eli Lilly, Merck Serono, MSD, Novartis, Ono Pharmaceutical, Pfizer, Sankyo Healthcare, and Taiho Pharmaceutical. RMJ has received honoraria and personal fees from and has participated in speakers' bureaus for Bristol-Myers Squibb. FC has been a consultant and participated in advisory boards for AstraZeneca, Bristol-Myers Squibb, MSD, Pfizer, Roche, and Takeda. FO has received research grants from Astellas Medivation, AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Celltrion and Roche; has participated in advisory boards for AstraZeneca, Bristol-Myers Squibb, Lilly, MSD, Pfizer, and Roche; and has received travel, accommodations, and expenses from AstraZeneca, Bristol-Myers Squibb, MSD, Pfizer, and Roche/Genentech. NN has received honoraria from AstraZeneca, Boehringer Ingelheim, Chugai Pharmaceutical, Kyowa Hakko Kirin, Eli Lilly Japan, MSD, Ono Pharmaceutical, Pfizer, and Taiho Pharmaceutical. DR-A has participated in speakers' bureaus for AstraZeneca, Bristol-Myers Squibb, MSD, Pfizer, and Roche. DM-S has received grants from AbbVie, Boehringer Ingelheim, Bristol-Myers Squibb, and Pfizer; has received personal fees from AbbVie, AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, MSD, Novartis, Pfizer, Roche, and Takeda; and participated in advisory boards and received travel expenses from AbbVie, AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, and Pfizer. CAT has participated in speakers' bureaus and advisory boards for Roche. FB has received personal fees from AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Eli Lilly Oncology, Merck, MSD, Novartis, Pfizer, Pierre Fabre, Roche, and Takeda; and reports institutional financial interest for clinical trials from AbbVie, ACEA, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol-Myers Squibb, Eisai, Eli Lilly Oncology, Genentech, Ignyta, Innate Pharma, Ipsen, Loxo, Medimmune, Merck, MSD, Novartis, Pierre Fabre, Pfizer, Sanofi-Aventis, and Takeda. GF has received personal fees for promotional speaking from Astellas Medivation, and Bristol-Myers Squibb; and has received non-branded presentation and speaking fees from Roche. AL, SC, YD, MK, GS, and WL are employees of Genentech and have ownership interest in Roche. MK was an employee of Genentech when this work was done, is a current employee of Bolt Biotherapeutics, and has a patent titled "Biomarkers and methods of treating PD-1 and PD-L1 related conditions" that is pending. MAS has received honoraria and research funding from Genentech and has participated in speakers' bureaus for Genentech. DS declares no competing interests.

Data sharing

Qualified researchers may request access to individual patient-level data through the clinical study data request platform (www.clinicalstudydatarequest.com). Further details about Roche's criteria for eligible studies are available at <https://clinicalstudydatarequest.com/Study-Sponsors/Study-Sponsors-Roche.aspx>. For further details about Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see: https://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm.

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